

# Self Supervised Learning

# What is Self-Supervised Learning?

Self-Supervised Learning (SSL) is a special type of representation learning that enables learning good data representation from unlabelled dataset.

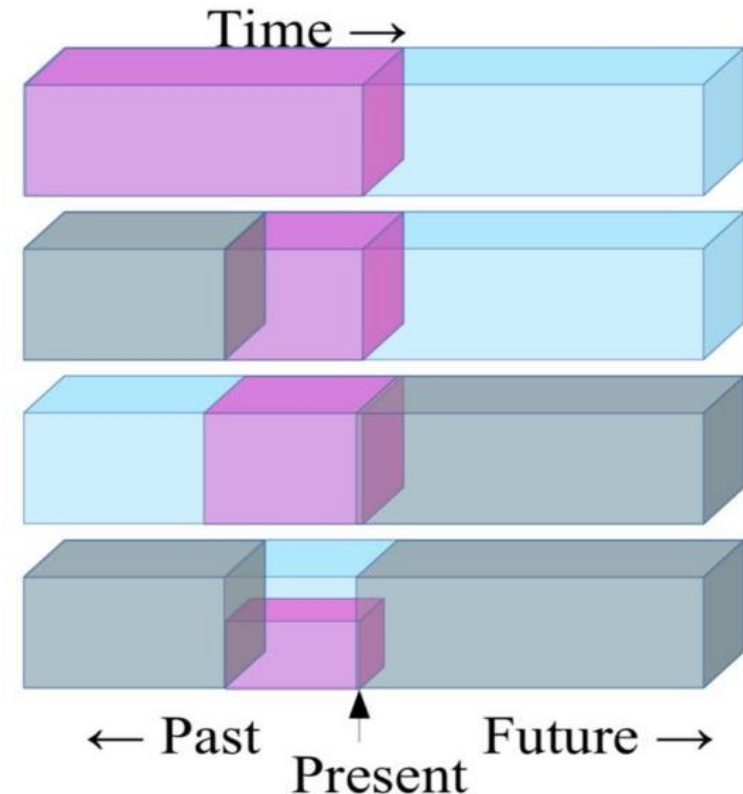
It is motivated by the idea of *constructing supervised learning tasks out of unsupervised datasets*. **Why?**

1. Data labeling is expensive and thus high-quality labeled dataset is limited.
2. Learning good representation makes it easier to transfer useful information to a variety of downstream tasks.
  - e.g. A downstream task has only a few examples.
  - e.g. Zero-shot transfer to new tasks.

# Self-Supervised Learning

General idea: pretend there is a part of the data you don't know and train the neural network to predict that.

- ▶ Predict any part of the input from any other part.
- ▶ Predict the **future** from the **past**.
- ▶ Predict the **future** from the **recent past**.
- ▶ Predict the **past** from the **present**.
- ▶ Predict the **top** from the **bottom**.
- ▶ Predict the **occluded** from the **visible**
- ▶ **Pretend there is a part of the input you don't know and predict that.**



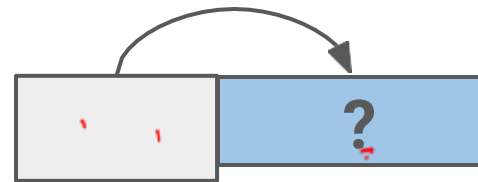
# Two Popular Ways

- ❑ Self-prediction
- ❑ Contrastive learning

# Methods for Framing Self-Supervised Learning Tasks

**Self-prediction:** Given an individual data sample, the task is to predict one part of the sample given the other part.

The part to be predicted pretends to be missing.

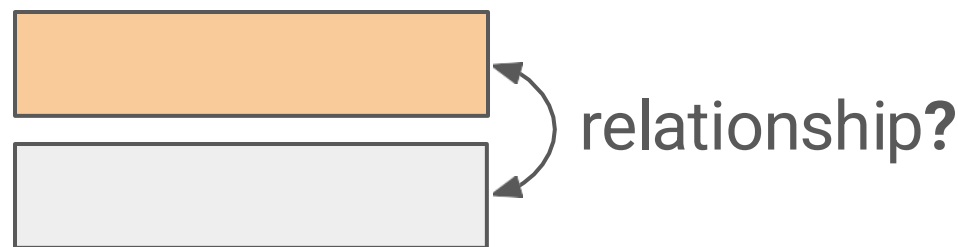


“Intra-sample” prediction

# Methods for Framing Self-Supervised Learning Tasks

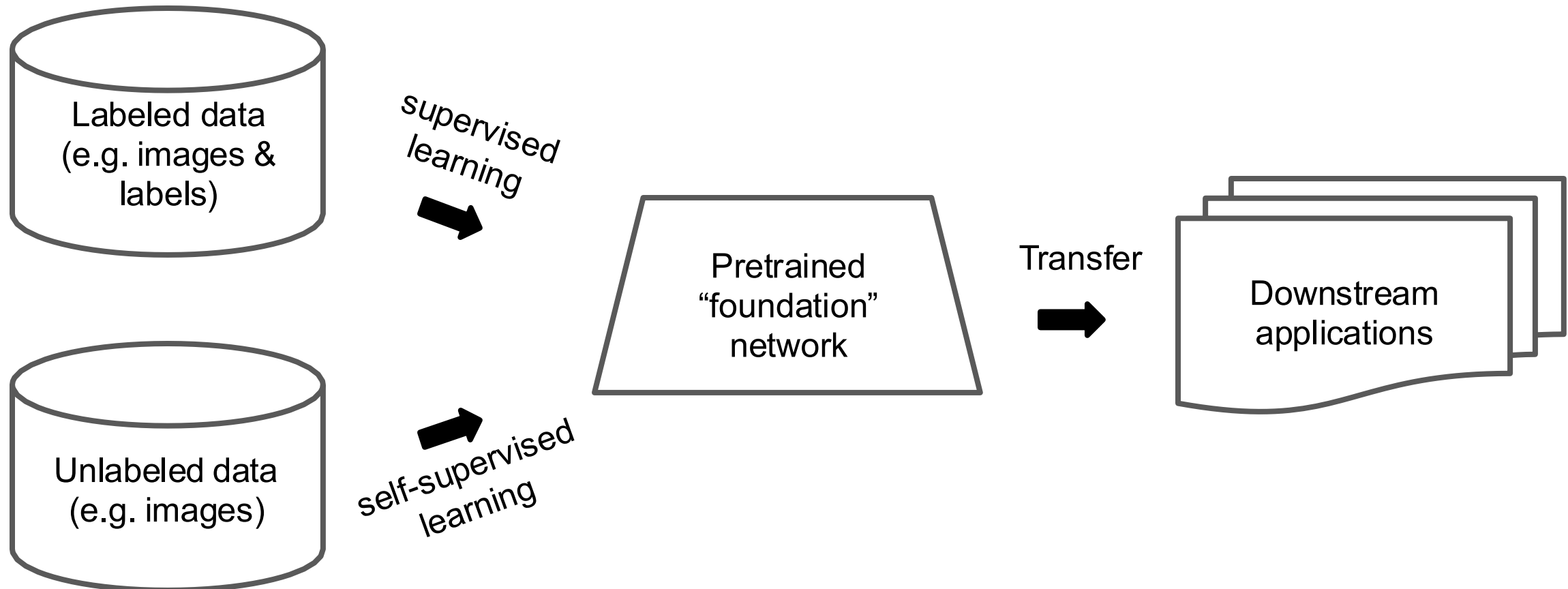
**Contrastive learning:** Given multiple data samples, the task is to predict the relationship among them.

The multiple samples can be selected from the dataset based on some known logics (e.g. the order of words / sentences) or fabricated by altering the original version.



“Inter-sample” prediction

# The paradigm of learning “foundation” models

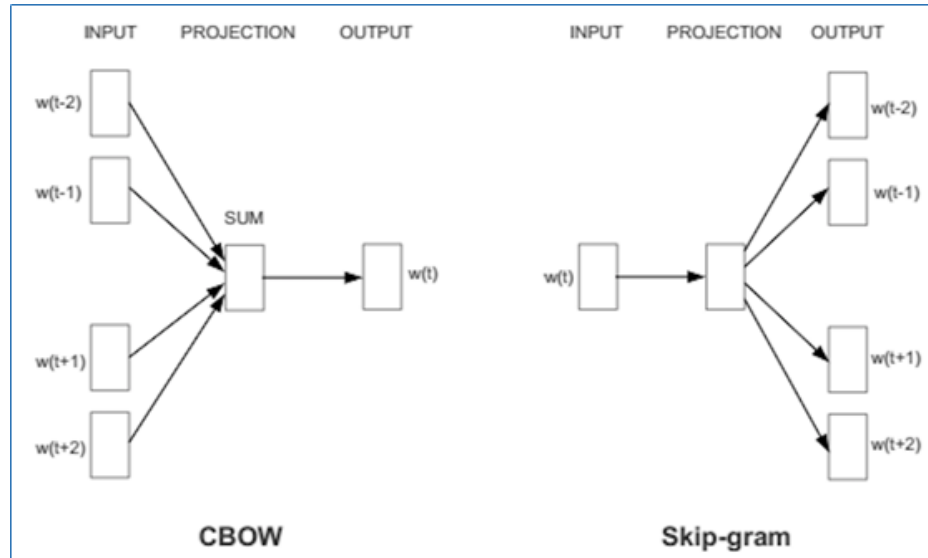


“Self-supervised learning” is “supervised learning” without specific task annotations.

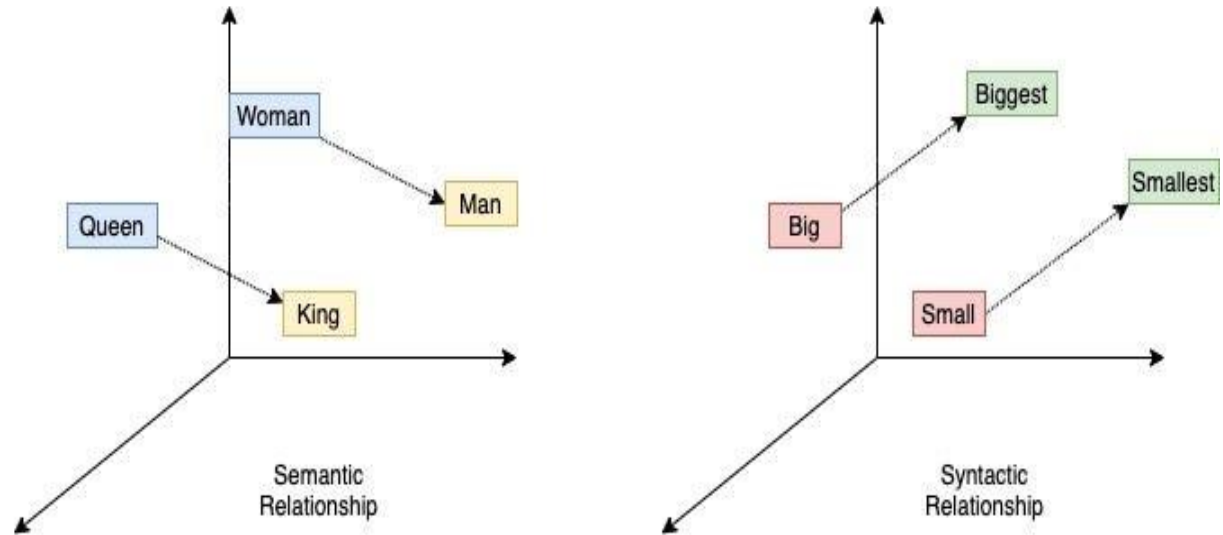
# Pre-Text Tasks

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# Word2Vec: Word Embedding



**Pretext Task of “Filling the Blank”**  
Training with no human supervision  
(Self-Supervised)



**King – Queen = Man - Woman**  
“Semantic” manipulation of text/words

# Examples from NLP

Self-supervised learning has driven the recent progress in the **Natural Language Processing** (NLP) field

- Models like ELMO, BERT, RoBERTa, ALBERT, Turing NLG, GPT-3 have demonstrated immense potential for automated NLP

Employing various pretext tasks for learning from raw text produced rich feature representations, useful for different downstream tasks

## Pretext tasks in NLP:

- Predict the center word given a window of surrounding words
  - The word highlighted with green color needs to be predicted



- Predict the surrounding words given the center word

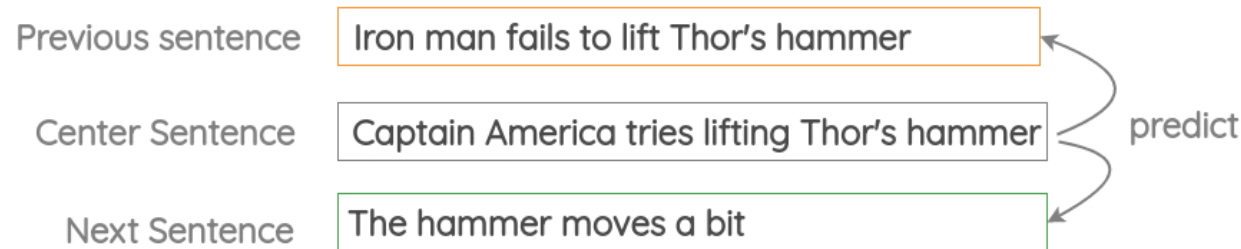
A quick brown fox jumps over the lazy dog

The words "quick", "brown", and "fox" are highlighted in green and enclosed in a box, indicating they are the words to be predicted given the center word "fox".

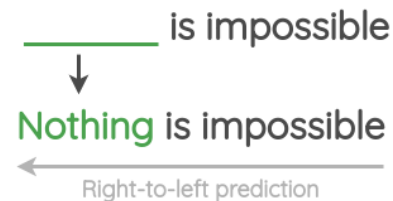
# Examples from NLP

## Pretext tasks in NLP:

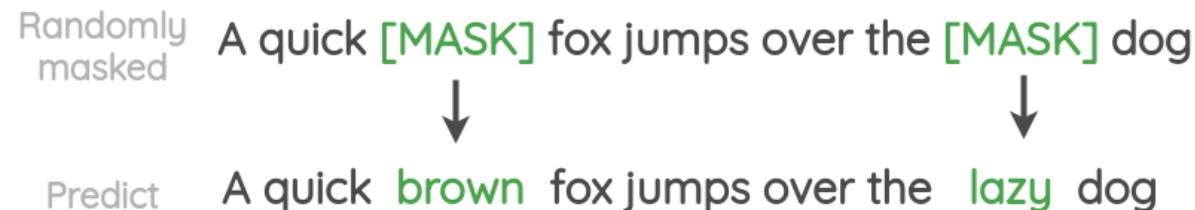
- From three consecutive sentences, predict the previous and the next sentence, given the center sentence



- Predict the previous or the next word, given surrounding words



- Predict randomly masked words in sentences



# Examples from NLP

## Pretext tasks in NLP:

- Predict if the ordering of two sentences is correct

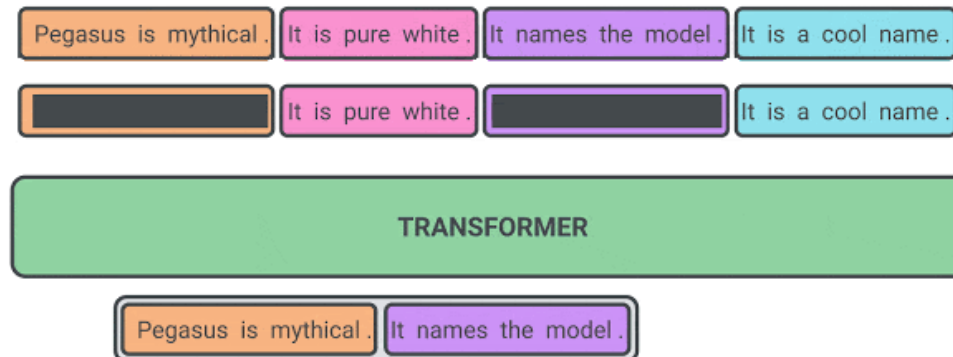
| Sentence 1         | Sentence 2                    | Next Sentence |
|--------------------|-------------------------------|---------------|
| I am going outside | I will be back in the evening | yes           |
| I am going outside | You know nothing John Snow    | no            |

- Predict the order of words in a randomly shuffled sentence

Finally I did Z. Then I did Y. I did X. Shuffle

I did X. Then I did Y. Finally I did Z. Recover

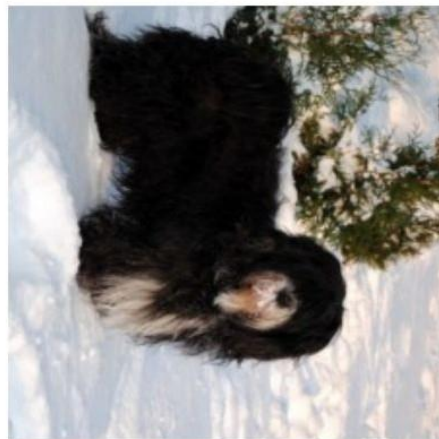
- Predict masked sentences in a document



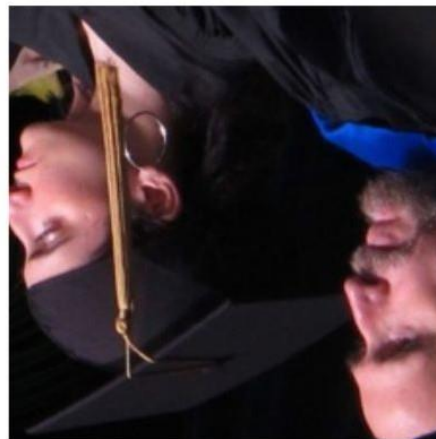
# Pretext task: predict rotations



90° rotation



270° rotation



180° rotation



0° rotation

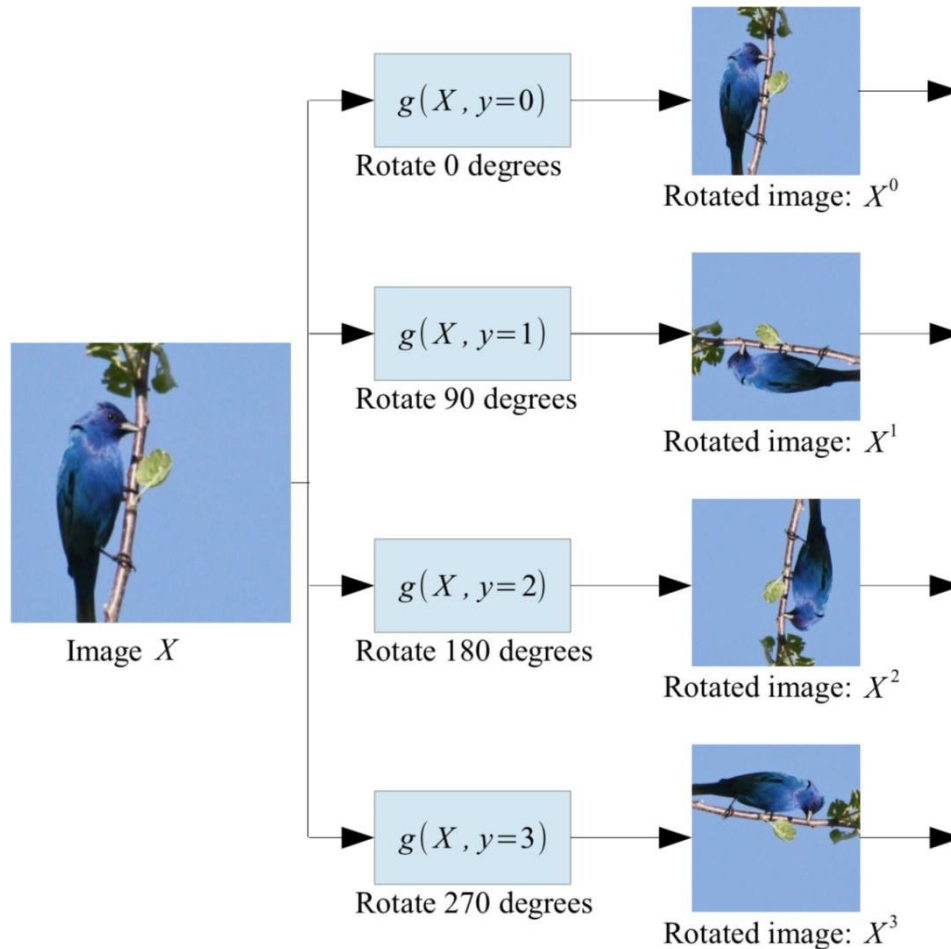


270° rotation

**Hypothesis:** a model could recognize the correct rotation of an object only if it has the “visual commonsense” of what the object should look like unperturbed.

(Image source: [Gidaris et al. 2018](#))

# Pretext task: predict rotations

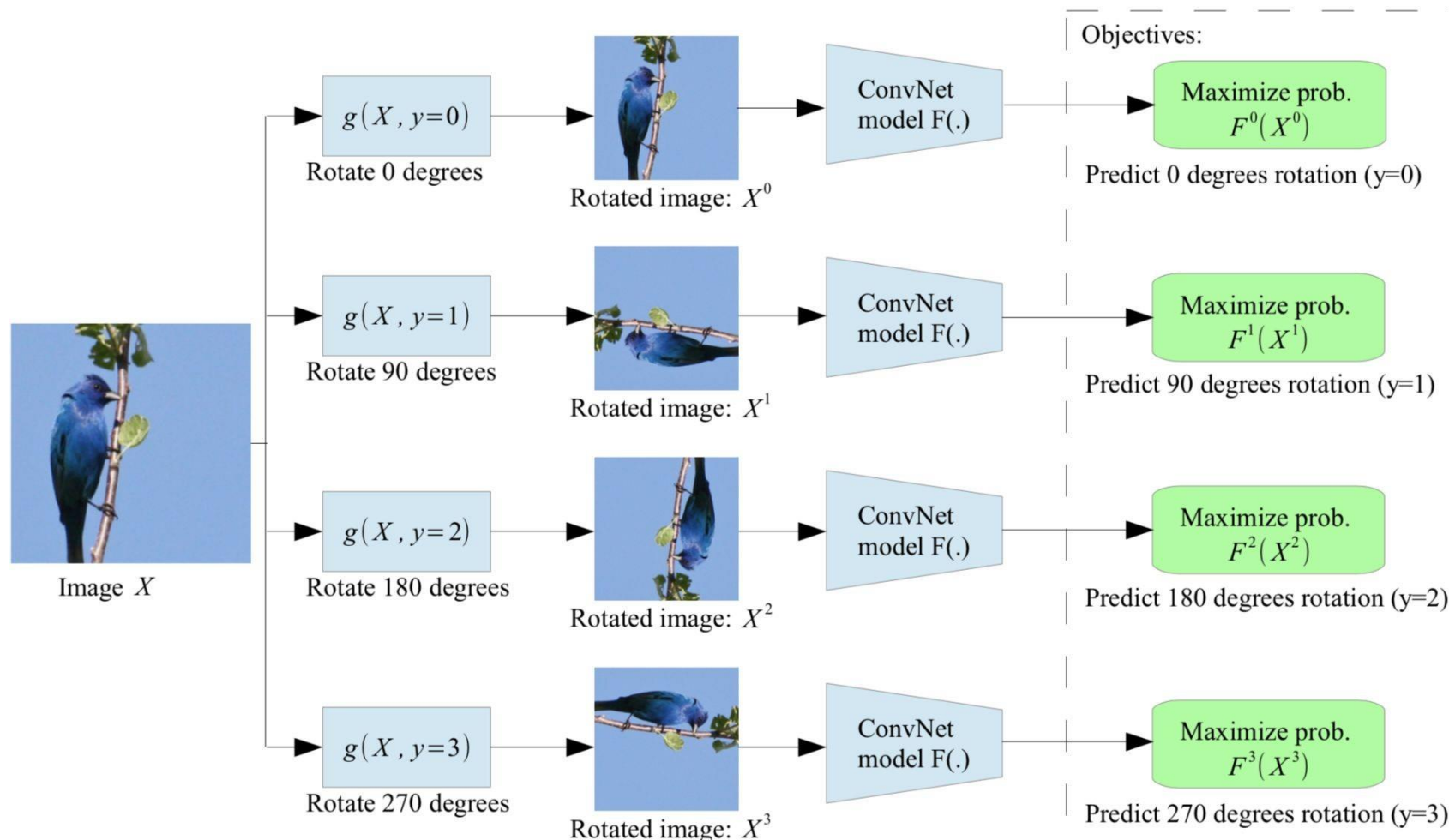


Self-supervised learning by rotating the entire input images.

The model learns to predict which rotation is applied (4-way classification)

(Image source: [Gidaris et al. 2018](#))

# Pretext task: predict rotations

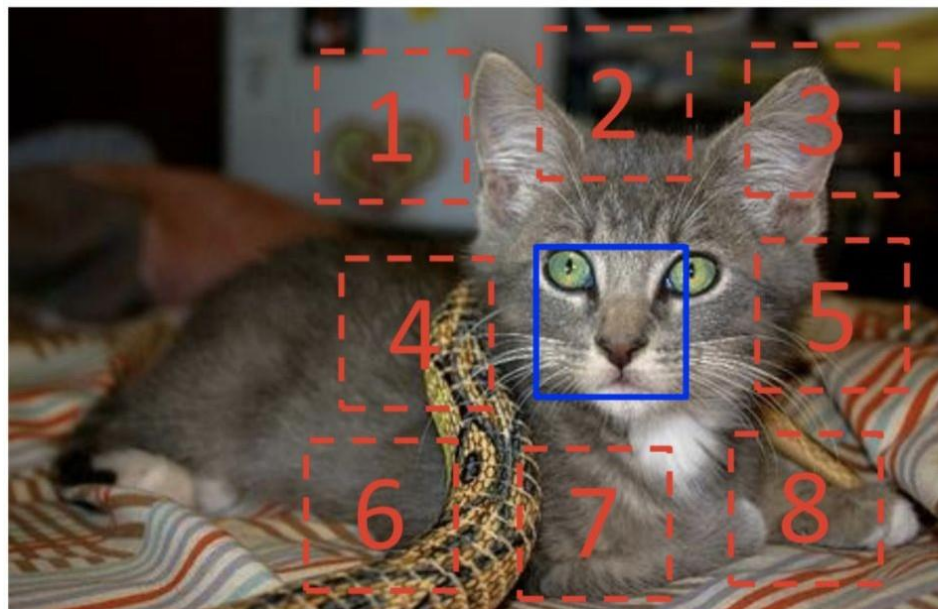


Self-supervised learning by rotating the entire input images.

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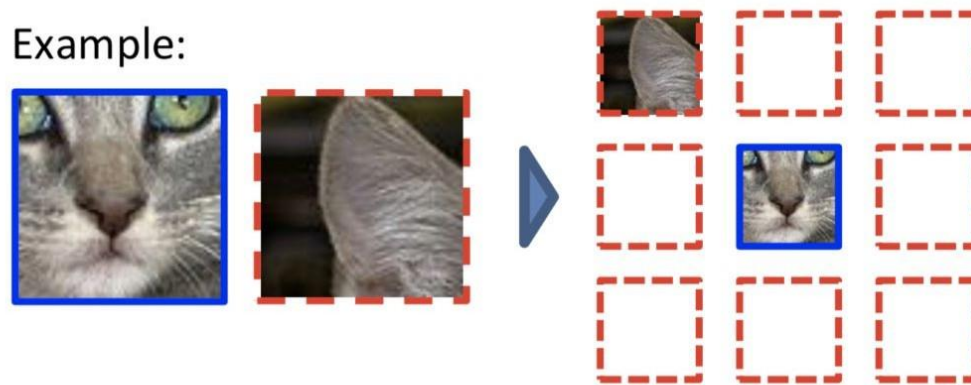
(Image source: [Gidaris et al. 2018](#))

# Pretext task: predict relative patch locations



$$X = (\text{cat face patch}, \text{cat ear patch}); Y = 3$$

Example:



Question 1:

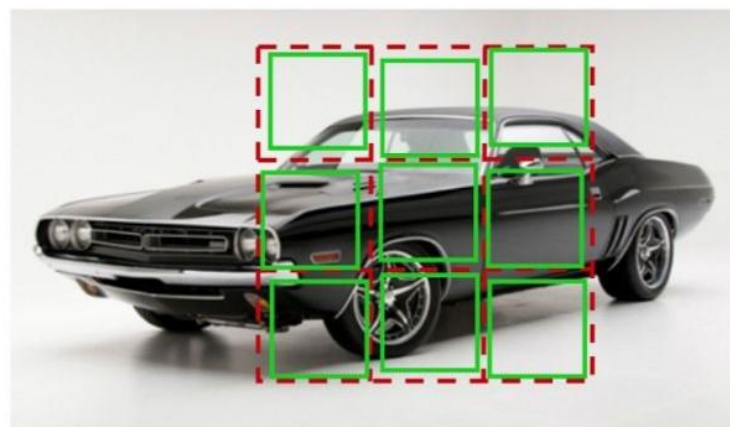


Question 2:



(Image source: [Doersch et al., 2015](#))

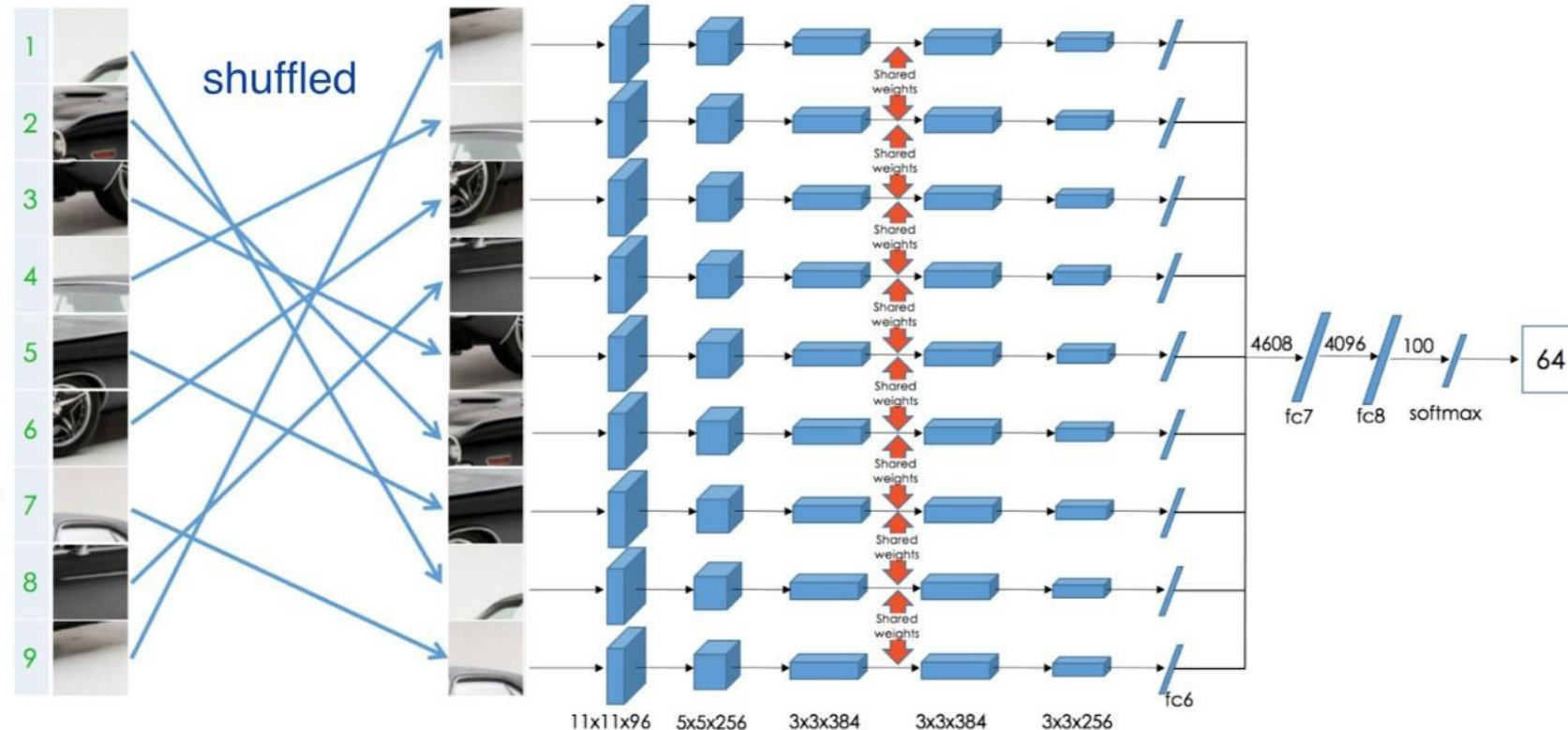
# Pretext task: solving “jigsaw puzzles”



Permutation Set

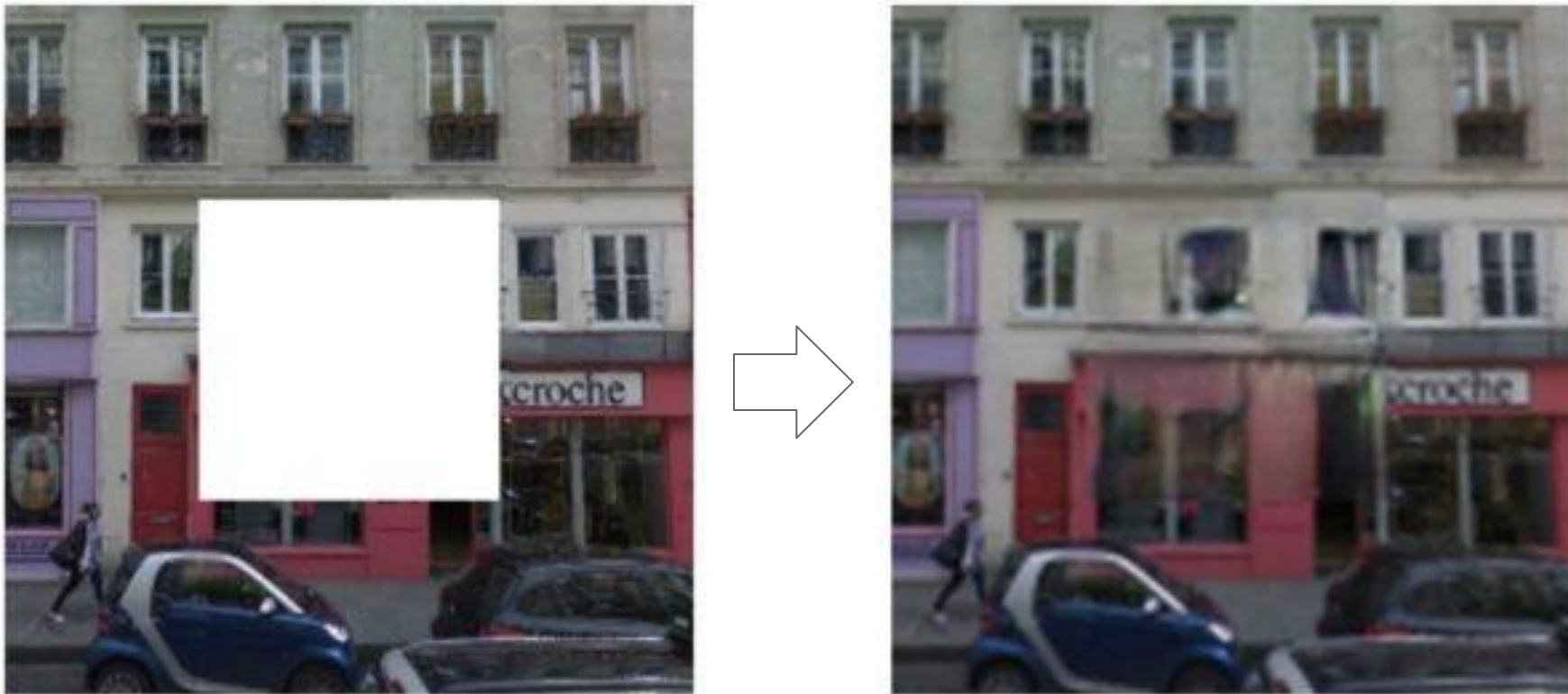
| index | permutation       |
|-------|-------------------|
| 64    | 9,4,6,8,3,2,5,1,7 |
|       |                   |
|       |                   |

Reorder patches according to the selected permutation



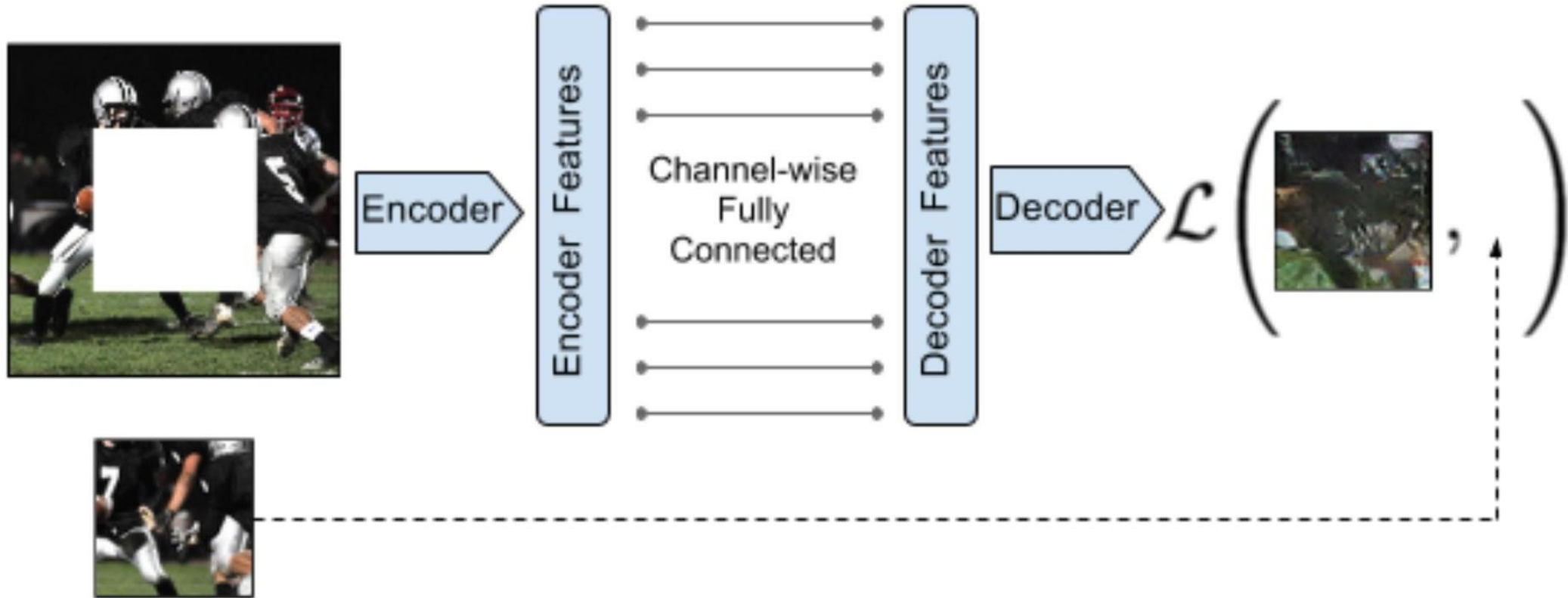
(Image source: [Noroozi & Favaro, 2016](#))

# Pretext task: predict missing pixels (inpainting,



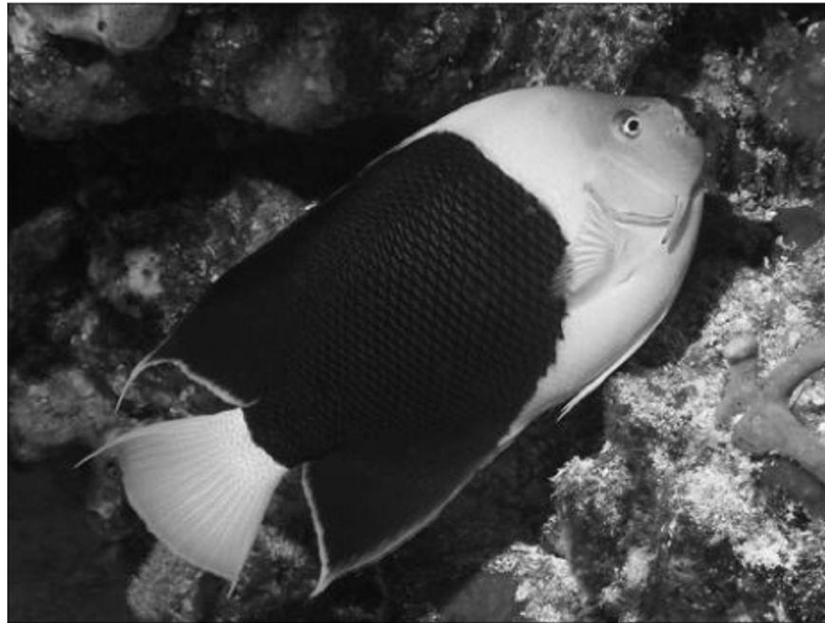
*Context Encoders: Feature Learning by Inpainting (Pathak et al., 2016)*

# Learning to inpaint by reconstruction



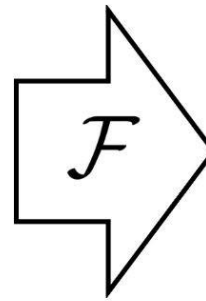
Learning to reconstruct the missing pixels

# Pretext task: image coloring



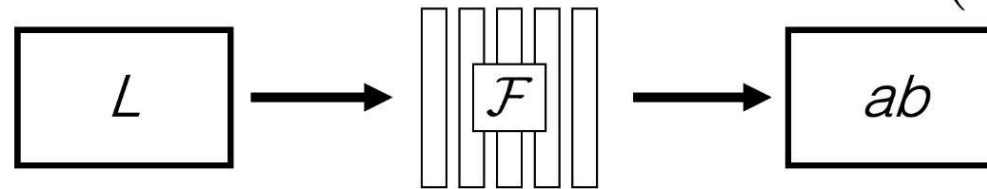
Grayscale image:  $L$  channel

$$\mathbf{X} \in \mathbb{R}^{H \times W \times 1}$$



Concatenate  $(L, ab)$  channels

$$(\mathbf{X}, \hat{\mathbf{Y}})$$



# Pretext task: video coloring

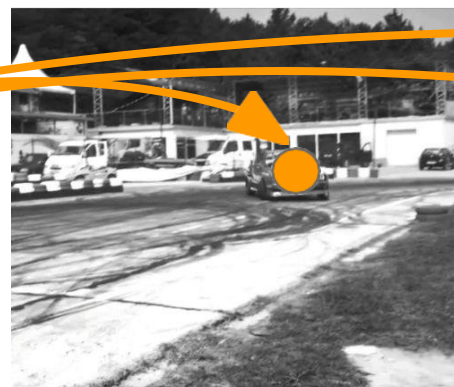
Idea: model the *temporal coherence* of colors in videos

reference frame

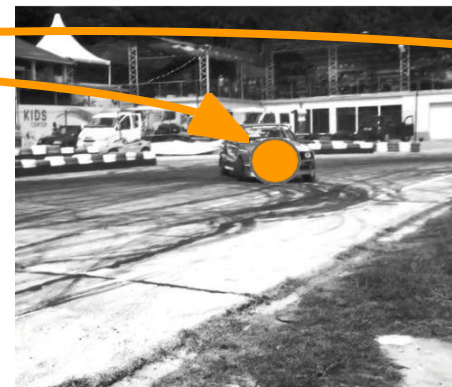


$t = 0$

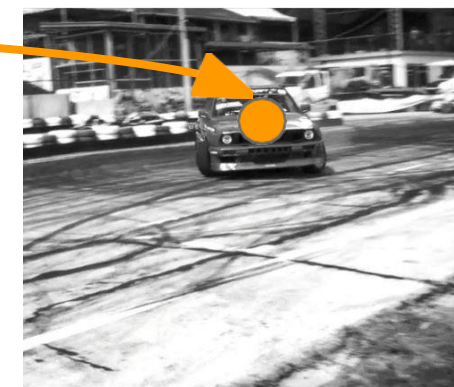
how should I color these frames?



$t = 1$



$t = 2$



$t = 3$

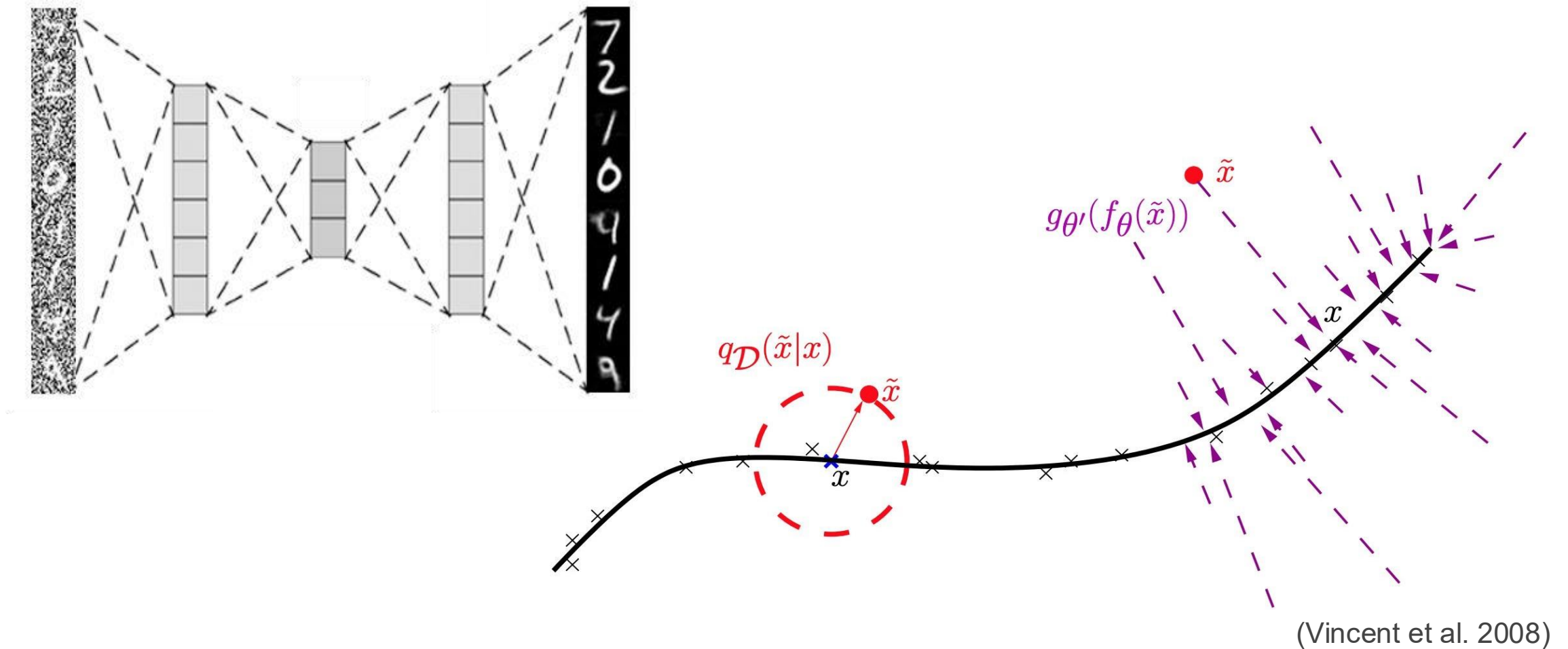
...

**Hypothesis:** learning to color video frames should allow model to learn to track regions or objects without labels!

Source: [Vondrick et al., 2018](#)

# Autoencoder: Self-Supervised Learning-Vision in

Denoising Autoencoder (Vincent et al. 2008)



# Pretext tasks from image transformations

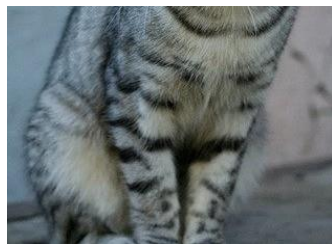
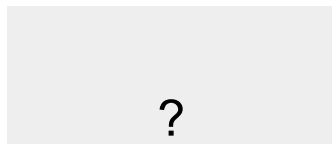
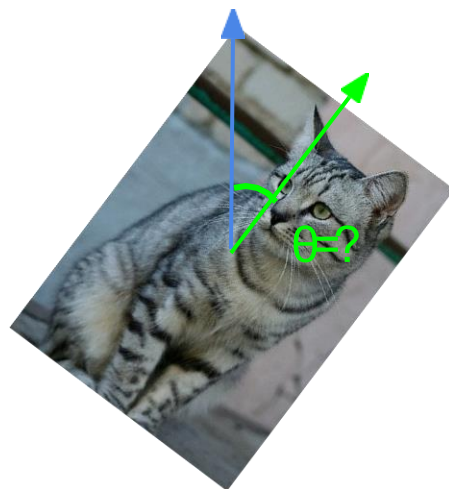
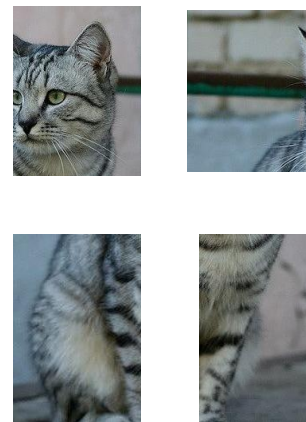


image completion



rotation prediction



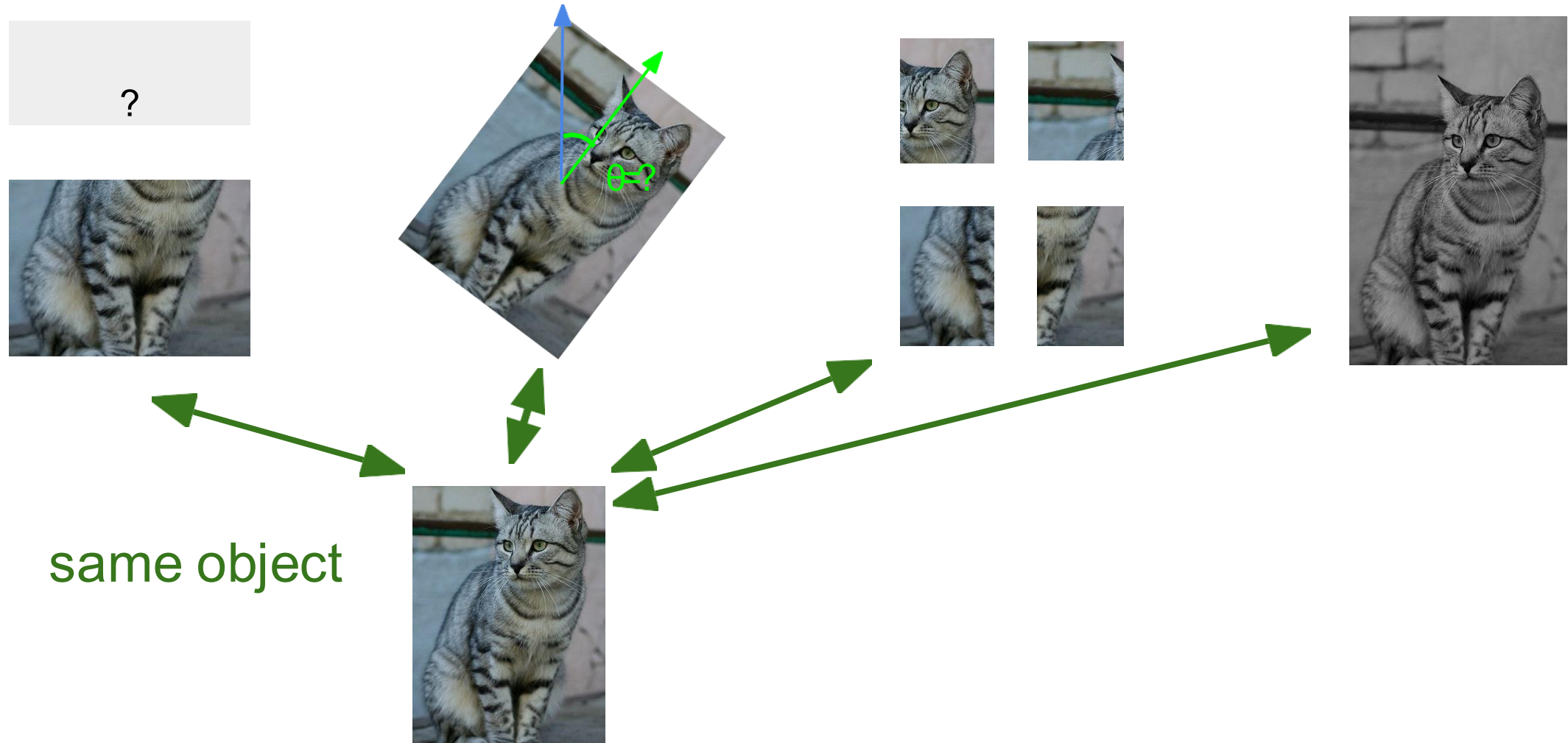
“jigsaw puzzle”



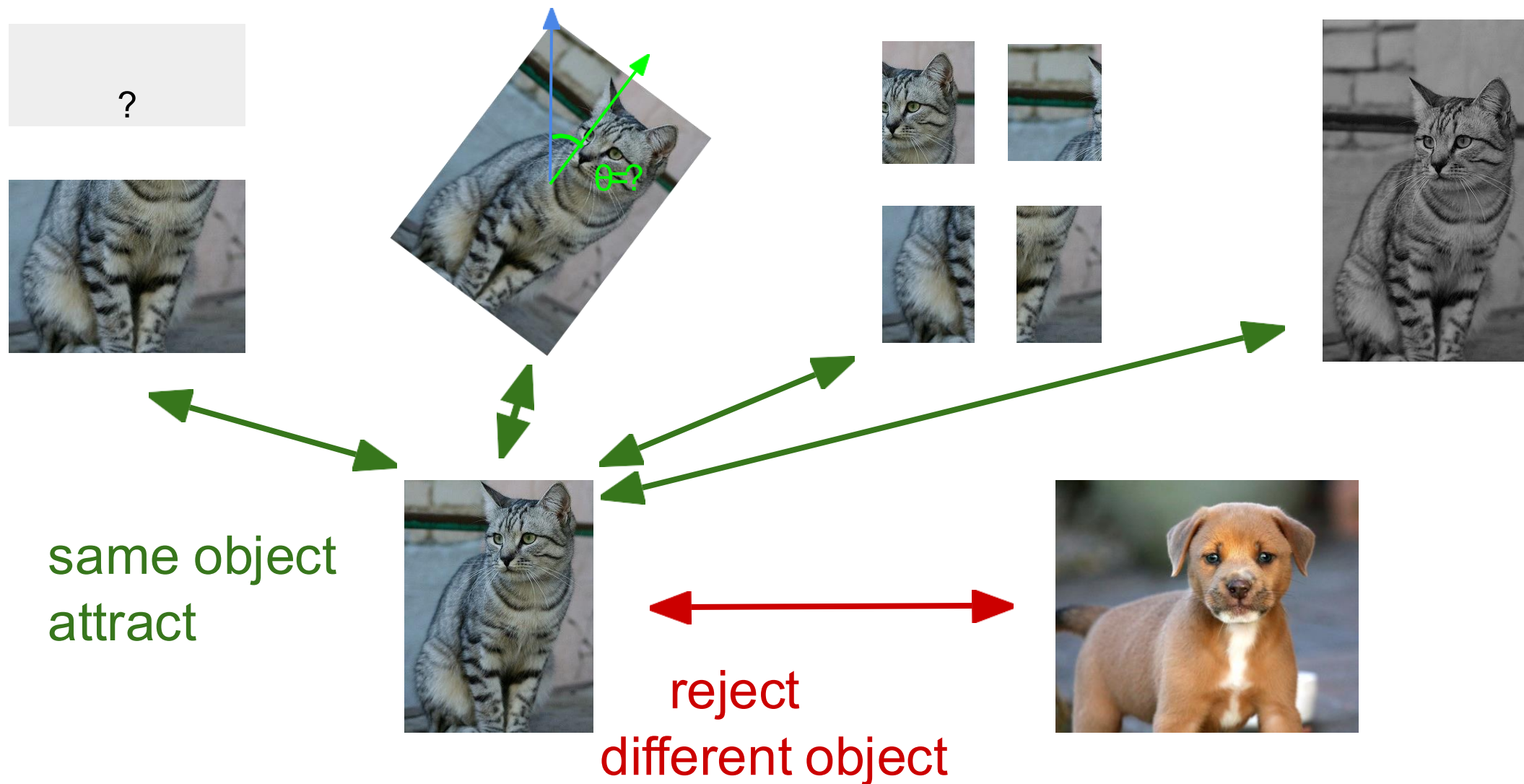
colorization

Learned representations may be tied to a specific pretext task! Can we come up with a more general pretext task?

# We know samples are same



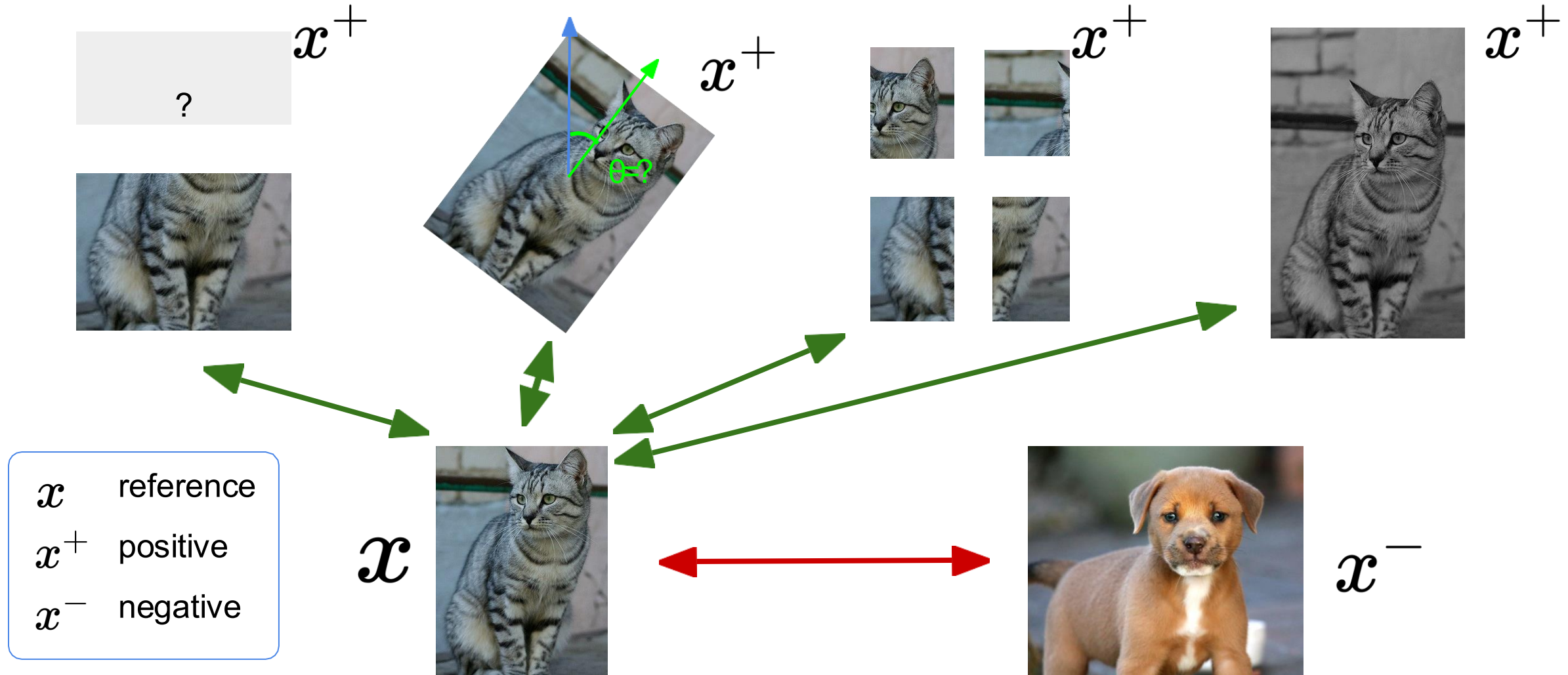
# We know samples are same and different



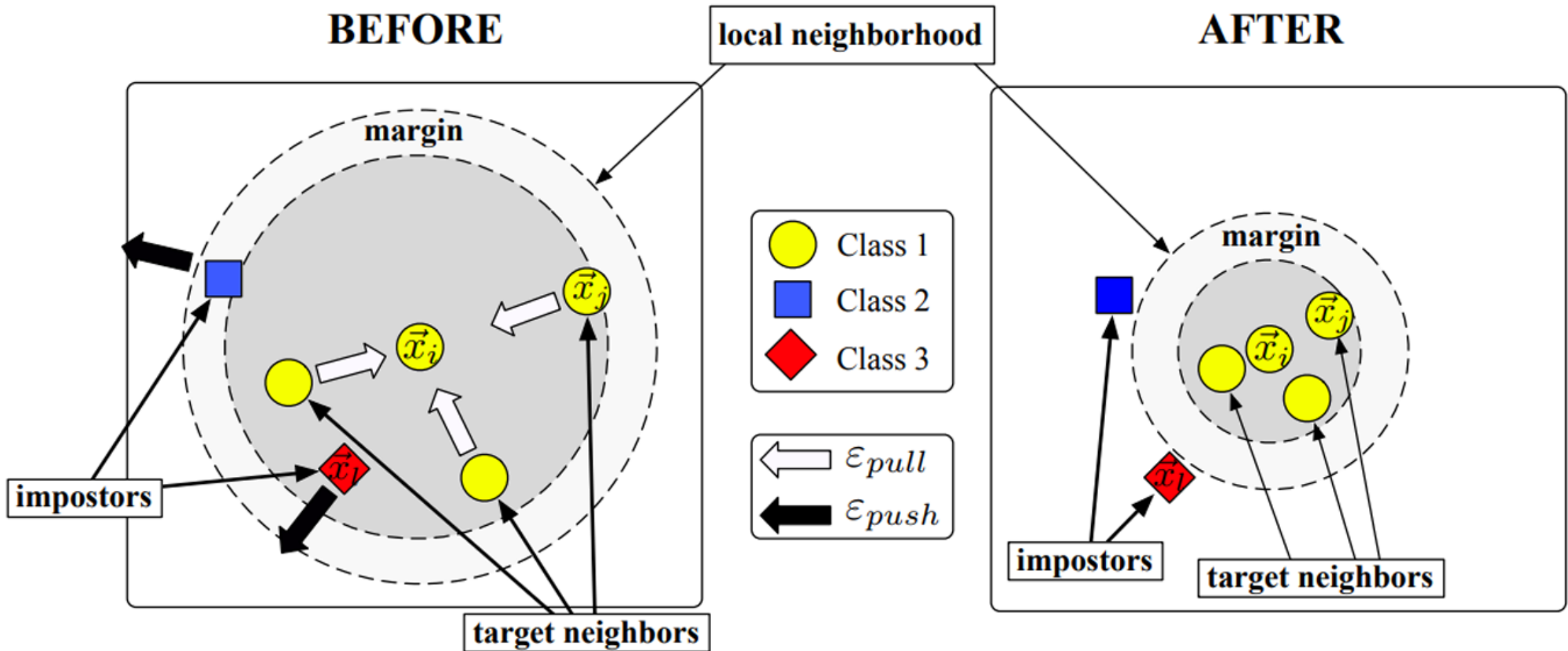
# Contrastive Learning

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# Contrastive Representation Learning

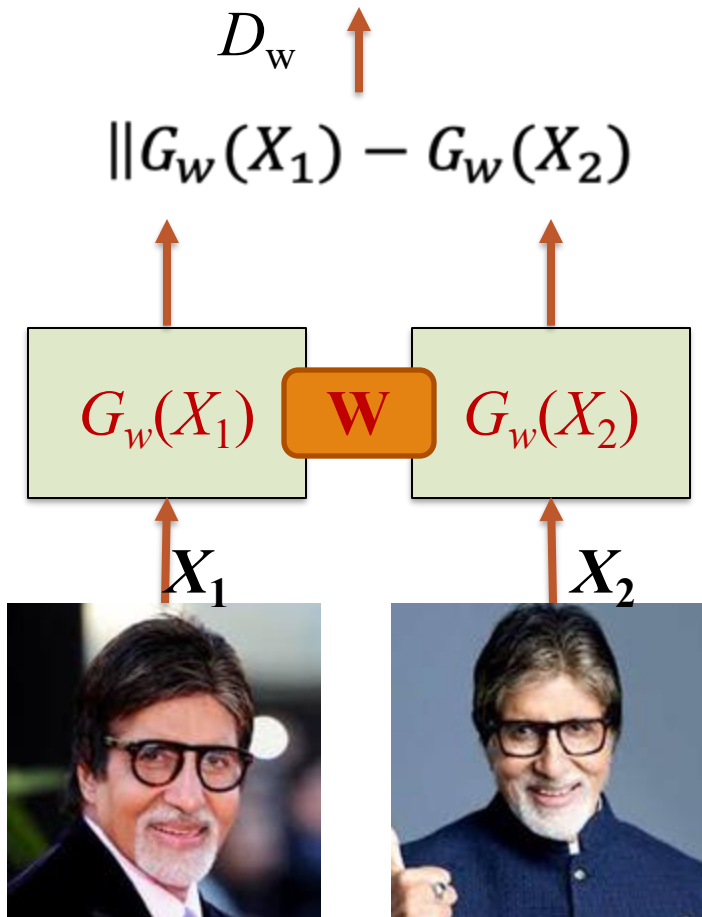


# Large Margin Nearest Neighbour(LMNN)

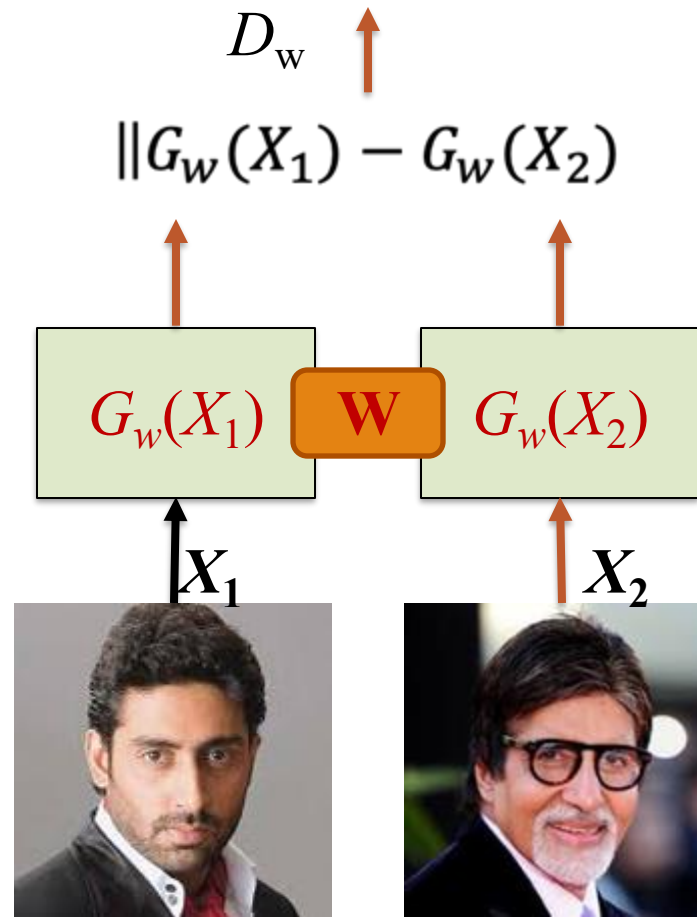


# Siamese Architecture/Loss

Make this smaller



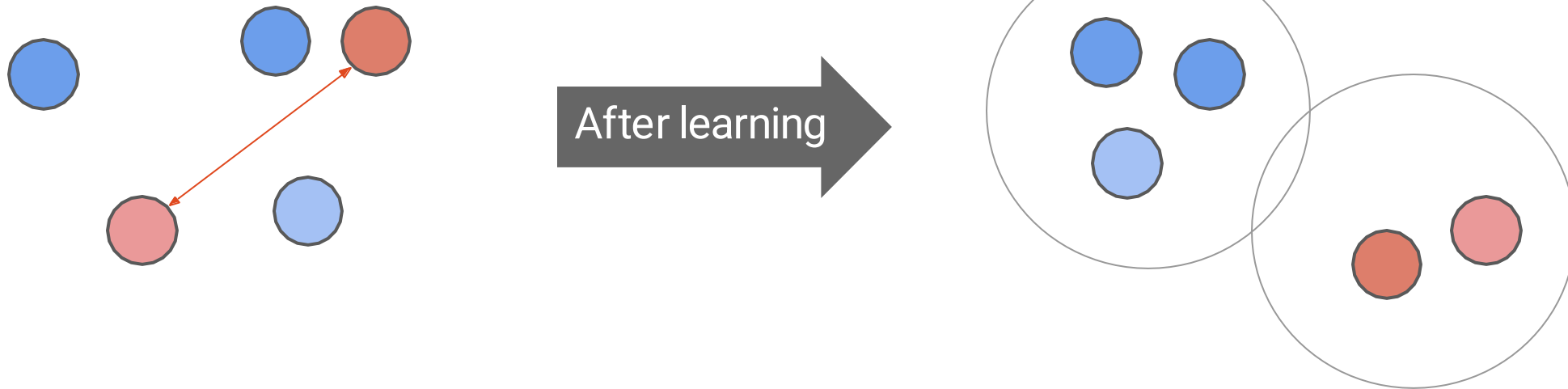
Make this larger



- Only pair-wise Labels
- Similarity Metric:  
 $D_w(X_1, X_2)$
- Have shared weights
- Training in batches

# Contrastive Learning

The goal of contrastive representation learning is to learn such an embedding space in which *similar* sample pairs stay *close* to each other while *dissimilar* ones are *far apart*.



# Contrastive Learning: Inter-Sample Classificati

The goal of contrastive representation learning is to learn such an embedding space in which *similar* sample pairs stay *close* to each other while *dissimilar* ones are *far apart*.

Given both similar (“positive”) and dissimilar (“negative”) candidates, to identify which ones are similar to the anchor data point is a classification task.

There are creative ways to construct a set of data point candidates:

- ❑ The original input and its distorted version
- ❑ Data that captures the same target from different views

# Contrastive Learning: Inter-Sample Classificati

Common loss functions:

- ❑ Contrastive loss (Chopra et al. 2005)
- ❑ Triplet loss (Schroff et al. 2015; FaceNet)
- ❑ Lifted structured loss (Song et al. 2015)
- ❑ Multi-class n-pair loss (Sohn 2016)
- ❑ Noise contrastive estimation (“NCE”; Gutmann & Hyvarinen 2010)
- ❑ InfoNCE (van den Oord, et al. 2018)
- ❑ Soft-nearest neighbors loss (Salakhutdinov & Hinton 2007, Frosst et al. 2019)

# Contrastive Learning: Inter-Sample Classificati

**Contrastive loss** (Chopra et al. 2005): Works with labelled dataset.

Encodes data into an embedding vector such that examples from the same class have similar embeddings and samples from different classes have different ones.

Given two labeled data pairs  $(\mathbf{x}_i, y_i)$  and  $(\mathbf{x}_j, y_j)$  :

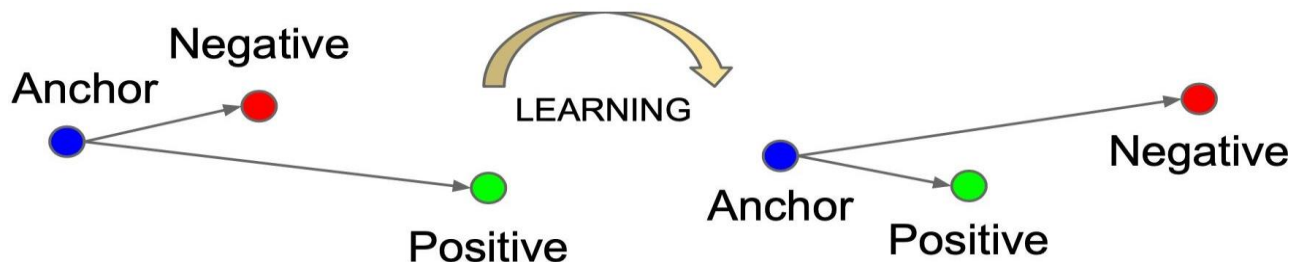
$$\mathcal{L}_{\text{cont}}(\mathbf{x}_i, \mathbf{x}_j, \theta) = \mathbb{1}[y_i = y_j] \underbrace{\|f_{\theta}(\mathbf{x}_i) - f_{\theta}(\mathbf{x}_j)\|_2^2}_{\text{minimize}} + \mathbb{1}[y_i \neq y_j] \max(0, \epsilon - \underbrace{\|f_{\theta}(\mathbf{x}_i) - f_{\theta}(\mathbf{x}_j)\|_2}_{\text{maximize}})^2$$

# Contrastive Learning: Inter-Sample Classificati

**Triplet loss** (Schroff et al. 2015): learns to minimize the distance between the anchor  $\mathbf{x}$  and positive  $\mathbf{x}^+$  and maximize the distance between the anchor  $\mathbf{x}$  and negative  $\mathbf{x}^-$  at the same time.

Given a triplet input  $(\mathbf{x}, \mathbf{x}^+, \mathbf{x}^-)$ ,

$$\mathcal{L}_{\text{triplet}}(\mathbf{x}, \mathbf{x}^+, \mathbf{x}^-) = \sum_{\mathbf{x} \in \mathcal{X}} \max(0, \|f(\mathbf{x}) - f(\mathbf{x}^+)\|_2^2 - \|f(\mathbf{x}) - f(\mathbf{x}^-)\|_2^2 + \epsilon)$$



(Schroff et al. 2015)

# Person identification



CUHK03 Data set



**True  
positive**



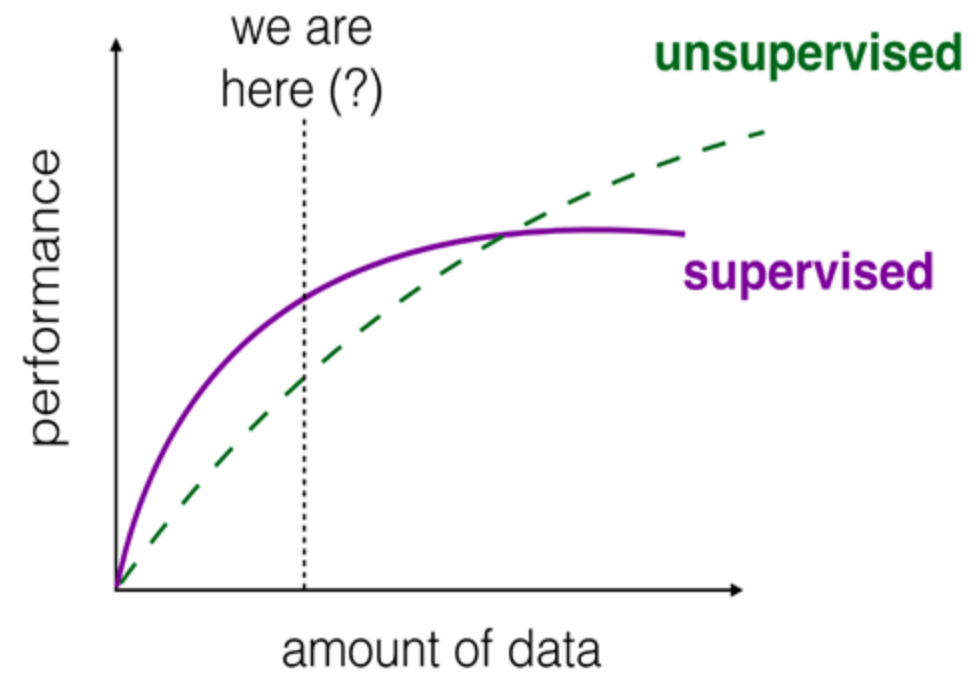
**True  
negative**

Ahmed, E, Jones, M. and Marks, T.K., 2015. An improved deep learning architecture for person re-identification. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition* (pp. 3908-3916).

# Summary

- ❑ Fully Supervised Learning is not practical in all situations
  - Availability of the annotation
  - Knowledge of the task ahead of time
- ❑ Self Supervised Learning provided an “unsupervised” learning
  - Pretraining
- ❑ Two main ways
  - Pretext Tasks
  - Contrastive Learning
- ❑ Success stories
  - Many in language
  - Impressive results in vision competing with Fully Supervised

# Unsupervised Learning of Face Representations



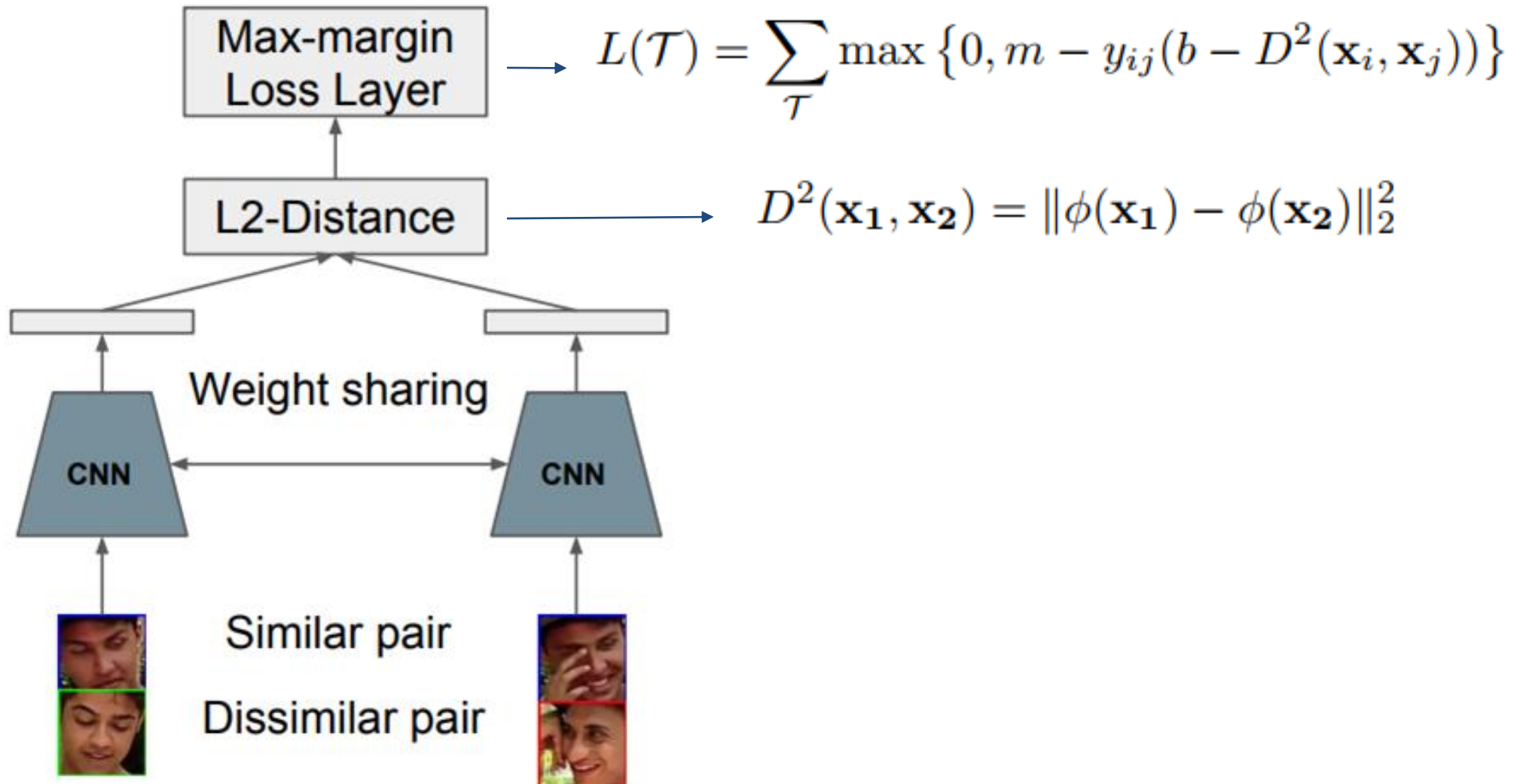


|         | Track 1 | Track 2 | Track 3 |
|---------|---------|---------|---------|
| Frame 1 |         |         |         |
| Frame 2 |         |         |         |
|         | ...     |         |         |
| Frame n |         |         |         |

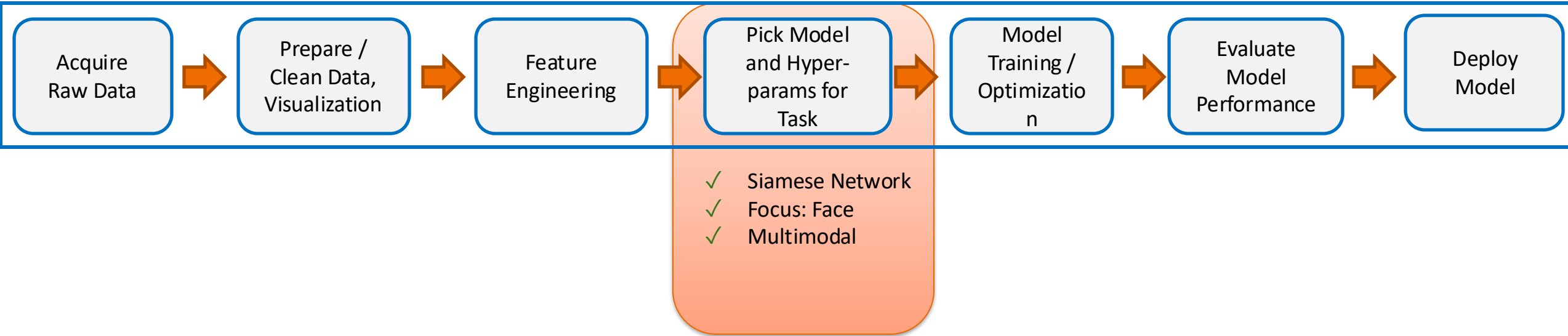
## Learn from YouTube

1. Multiple faces in the same frame must belong to different people
2. The same face tracked across multiple frames belongs to the same identity

# Model



# Summary



# Thanks!!

## Questions?

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